

Assignment2 Micro Economics B

zts900, chs380

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1. Straffespark

1.1

We have that, both the goalkeeper and the kicker observe the kicker's type. We therefore solve two separate 3×3 normal-form games: One game where the kicker is left-footed and one game where the kicker is right-footed.

Each game is modeled as a zero-sum game, where the goalkeeper's payoff is the probability of making a save, and the kicker's payoff is the complement (probability of scoring).

We compute the mixed strategy Nash equilibria using the support enumeration method.

The results are:

- Left-footed kicker:
 - Goalkeeper's strategy: $(L, C, R) = (0.468, 0.108, 0.424)$
 - Kicker's strategy: $(L, C, R) = (0.426, 0.242, 0.332)$
- Right-footed kicker:
 - Goalkeeper's strategy: $(L, C, R) = (0.727, 0.075, 0.197)$
 - Kicker's strategy: $(L, C, R) = (0.432, 0.233, 0.335)$

Interpretation: The optimal strategies reflect the type-dependent tendencies: right-footed kickers tend to be more accurate when shooting left, while left-footed kickers are

stronger to the right. The goalkeeper adjusts accordingly by placing more probability mass on the side that is most likely to be attacked by the given type.

1.2

In this setting, only the kicker observes their own type (left-footed or right-footed), while the goalkeeper must commit to a single strategy based on the known type distribution:

$$\Pr(\theta = \theta_V) = 0.11 \text{ (left-footed kicker)}$$

$$\Pr(\theta = \theta_H) = 0.89 \text{ (right-footed kicker)}$$

We model this situation as a Bayesian game with incomplete information. The goalkeeper maximizes their expected payoff (probability of saving) given the type distribution and the assumption that each kicker type plays a best response to the goalkeeper's strategy.

The resulting Bayesian Nash Equilibrium is:

- **Goalkeeper's strategy:** $(L, C, R) = (0.727, 0.075, 0.197)$
- **Left-footed kicker's strategy:** $(L, C, R) = (0, 0, 1)$
- **Right-footed kicker's strategy:** $(L, C, R) = (0, 0, 1)$

Interpretation: Given that right-footed kickers are in the majority, the goalkeeper's strategy is heavily influenced by defending against them. Both kicker types then respond by choosing the same action (kicking right), which effectively conceals the kicker's type and eliminates any advantage the goalkeeper might gain from statistical inference.

This highlights a typical result in Bayesian games: players with private information may mimic each other to remain unpredictable, thereby neutralizing the informational advantage of their opponent.

1.3

In this case, we assume that the distribution of kicker types is reversed compared to the previous scenario:

Left-footed kicker: 89%

Right-footed kicker: 11%

We solve the Bayesian game under this new information structure. The goalkeeper does not observe the kicker's type and must choose a single strategy based on the type distribution. Each kicker type, however, knows its own type and best-responds to the goalkeeper's mixed strategy.

The resulting Bayesian Nash Equilibrium is: Goalkeeper's strategy: $(L, C, R) = (0.468, 0.108, 0.424)$

Left-footed kicker's strategy: $(L, C, R) = (1, 0, 0)$

Right-footed kicker's strategy: $(L, C, R) = (1, 0, 0)$

Interpretation: The left-footed type (now in the majority) dictates the goalkeeper's optimal strategy, leading both types to adopt the same action. This strategic alignment allows the minority type to "hide" behind the majority, making it harder for the goalkeeper to exploit type-specific weaknesses.

Thus, even though types differ in ability and preferred direction, the optimal behavior under incomplete information results in both types choosing the same pure strategy, which minimizes the goalkeeper's ability to respond effectively.

1.4

We compare the strategies in three situations:

Situation	Goalkeeper's strategy (L, C, R)	Attacker's strategy (L, C, R)
Venstrebenet, kendt type	(0.468, 0.108, 0.424)	(0.426, 0.242, 0.332)
Højrebenet, kendt type	(0.727, 0.075, 0.197)	(0.432, 0.233, 0.335)
Ukendt type, 11% venstre	(0.727, 0.075, 0.197)	Begge typer: (0, 0, 1)
Ukendt type, 89% venstre	(0.468, 0.108, 0.424)	Begge typer: (1, 0, 0)

Table 1: Strategier i forskellige informations- og fordelingsscenarier

When the attacker is in the minority, it is strategically wise to mimic the behavior of the majority in order to conceal their type. The goalkeeper must defend based on the most likely type, which allows the rare type to adapt and avoid being exploited.

In Bayesian equilibria, both types often end up playing the same action, which removes the goalkeeper's ability to optimize against specific types.

It is advantageous to be among those in the minority, as the majority determines the goalkeeper's strategy, and the minority can hide by playing like the majority.

2. Dynamisk Priskonkurrence

2.1

We consider the one-shot stage game where both firms simultaneously choose prices to maximize their profits. With $\alpha = 0$, the utility index simplifies to $v_i = 1 - p_i$, and the demand function becomes:

$$D_i(p_i, p_j) = \frac{e^{1-p_i}}{e^{1-p_i} + e^{1-p_j}}.$$

The profit function of firm i is given by:

$$\pi_i(p_i, p_j) = D_i(p_i, p_j) \cdot p_i.$$

Assuming symmetry ($p_1 = p_2 = p$), we solve:

$$\max_p \pi_i(p, p).$$

Using numerical optimization, we find that the symmetric Nash equilibrium price is:

$$p^* \approx 1.5671.$$

This is the static Nash equilibrium in pure strategies under the Cournot parameterization ($\alpha = 0$).

2.2

We now consider the price that a static cartel would choose in order to maximize total industry profit, assuming both firms set the same price p .

The joint profit function is:

$$\pi_{\text{total}}(p) = \pi_1(p, p) + \pi_2(p, p).$$

Using the demand specification with $\alpha = 0$, and solving numerically, we find that the profit-maximizing symmetric cartel price is:

$$\bar{p} \approx 1.8526.$$

This price is higher than the Nash equilibrium price from question 2.1, as expected under collusion, since firms internalize the effect of their pricing on total industry profit.

2.3

To propose how the cartel should divide its total profit between the two firms, we apply Nash's bargaining solution from bargaining theory.

We assume that the two firms negotiate over the distribution of the total profit generated by the cartel. Each firm's outside option is given by the profit it would obtain in the absence of a cartel, (the profit under the static Nash equilibrium derived in question 2.1.) Furthermore, we assume symmetric bargaining power and, if relevant, a common discount factor $\delta \in (0, 1)$.

Nash Bargaining Solution. The Nash bargaining solution prescribes that the profit allocation (x_1, x_2) solves:

$$\max_{x_1 + x_2 = \pi_{\text{cartel}}} (x_1 - v_1)(x_2 - v_2),$$

where π_{cartel} is the total cartel profit, and v_1, v_2 are the outside options of firm 1 and firm 2, respectively.

This solution ensures that each firm receives at least what it could obtain outside the cartel, and the remaining profit is split evenly under the assumption of symmetric bar-

gaining power:

$$x_1 = v_1 + \frac{1}{2}(\pi_{\text{cartel}} - v_1 - v_2), \quad x_2 = v_2 + \frac{1}{2}(\pi_{\text{cartel}} - v_1 - v_2).$$

The Nash bargaining solution offers a fair and efficient way to distribute cartel profits. It respects both firms' fallback positions and incentivizes cooperation by ensuring that neither firm is worse off by participating in the cartel.

2.4

We consider a trigger strategy where both firms begin by playing the cartel price $\bar{p}$ and continue to do so as long as no firm deviates. If any firm deviates, the punishment is to revert permanently to the static Nash equilibrium price p^* .

To find the minimum discount factor δ such that this strategy profile is a subgame perfect Nash equilibrium, we compare the lifetime profit from cooperating with the cartel versus deviating unilaterally in a single period and then receiving Nash profits forever after.

Lifetime profit from cooperation:

$$V_{\text{coop}} = \frac{\pi_C}{1 - \delta},$$

where π_C is the per-period profit under the cartel price. Based on our computations:

$$\pi_C \approx 0.4263$$

Lifetime profit from deviation:

$$V_{\text{dev}} = \pi_D + \frac{\delta \pi_N}{1 - \delta},$$

where π_D is the one-time profit from undercutting the cartel, and π_N is the profit under the Nash equilibrium:

$$\pi_D \approx 0.4480, \quad \pi_N \approx 0.4164$$

Incentive compatibility condition: To deter deviation, we require:

$$V_{\text{coop}} \geq V_{\text{dev}} \implies \delta \geq \frac{\pi_D - \pi_C}{\pi_D - \pi_N}$$

Result: Plugging in the values yields:

$$\delta_{\min} \approx \frac{0.4480 - 0.4263}{0.4480 - 0.4164} \approx 0.6870$$

This implies that firms must value future profits sufficiently for collusion to be sustainable.

Implied annual interest rate: Using the relation $\delta = \frac{1}{1+r}$, we solve for r :

$$r = \frac{1}{\delta} - 1 \approx 45.55\%$$

An annual interest rate of approximately 45.55% is very high and significantly above current market interest rates. Therefore, under current financial conditions, such collusion would likely not be sustainable unless firms are extremely patient or heavily penalized for price competition.

2.5

We now consider the case where $\alpha \rightarrow \infty$, meaning the products are perfect substitutes. In this case, the model approaches the pure Bertrand competition framework, where consumers always choose the firm offering the lowest price.

In the original trigger strategy, both firms agree to play the cartel price $\bar{p}$ and revert to the Nash equilibrium price p^* in case of deviation. However, when $\alpha \rightarrow \infty$, even an infinitesimal price undercut (e.g., $p = \bar{p} - \varepsilon$) allows a firm to capture the entire market demand, making the deviation extremely profitable.

$$\pi_C = \frac{\bar{p}}{2} \quad \text{vs.} \quad \pi_D \approx \bar{p}$$

Moreover, under perfect substitutes, the Nash equilibrium price is $p^* = 0$, leading to zero

profit in the punishment phase. Thus, while the punishment is severe, it may not be credible, and the temptation to deviate is too high to sustain the equilibrium with the original trigger strategy.

We propose a grim trigger strategy where:

- Play the cartel price $\bar{p}$ as long as no firm has deviated.
- If a deviation occurs, both firms revert permanently to the zero-profit Bertrand outcome, i.e., $p = 0$.

To sustain collusion, the discounted value of cooperation must exceed the one-shot gain from deviation:

$$V_{\text{coop}} = \frac{\bar{p}}{2(1 - \delta)} \geq V_{\text{dev}} = \bar{p}$$

Solving for δ :

$$\frac{1}{2(1 - \delta)} \geq 1 \Rightarrow 1 - \delta \leq \frac{1}{2} \Rightarrow \delta \geq \frac{1}{2}$$

When products are perfect substitutes, firms face extreme competition pressure, and deviations are highly profitable. The original trigger strategy fails to sustain cooperation. However, under a grim trigger strategy with a sufficiently high discount factor $\delta \geq \frac{1}{2}$, collusion can be supported as a subgame perfect Nash equilibrium.

2.6

We now consider the case where $\alpha = 1$, corresponding to an intermediate degree of product differentiation. In this setting, the model is neither a pure Bertrand competition (as in $\alpha \rightarrow \infty$), nor a fully differentiated Cournot-type environment (as in $\alpha = 0$).

In question 2.4, we proposed a trigger strategy where firms cooperate at the cartel price $\bar{p}$, and revert to the Nash price p^* if any firm deviates. This strategy was sustainable for high enough discount factors δ , because the gain from deviation was limited and the punishment credible.

However, when $\alpha = 1$, products are closer substitutes, and undercutting becomes more attractive: the deviating firm captures a larger market share at the expense of the other. As a result, the one-shot deviation profit π_D becomes significantly higher than the cooperative profit π_C , while the punishment profit π_N is not sufficiently low to deter deviation. Consequently, no reasonable discount factor δ can make this strategy credible, which means it is no longer a subgame perfect equilibrium.

In question 2.5, we proposed a grim trigger strategy where any deviation results in a reversion to zero-profit Bertrand pricing ($p = 0$). This strategy relies on the assumption that the Nash equilibrium yields zero profits.

However, at $\alpha = 1$, product differentiation still exists, and thus the Nash equilibrium price p^* is strictly positive and leads to strictly positive profits ($\pi_N > 0$). This makes the punishment less severe and significantly weakens its deterrent effect. The deviation profit π_D may still exceed the present value of cooperation due to imperfect punishment, and hence the strategy fails to satisfy the incentive compatibility condition.

At $\alpha = 1$, the model lies in a “gray zone” where neither the Cournot-style nor the Bertrand-style trigger strategy is sufficient to sustain cooperation. In both cases, the threat of punishment is not strong enough to deter profitable deviation, making both strategies fail the subgame perfection requirement.

3 Ebay

3.A.1

Finding BNE for the auction The Bayesian-Nash equilibrium BNE in a second price sealed bid auction SPSB, with IPV values $v_i \sim U(0, 1)$. There are no profitable deviations from bidding the valuation regardless of what the other player does, and hence this is a BNE, no bidder can raise expected surplus by deviating. For every bidder i we have BNE equilibrium strategy profile:

$$b_i(v_i) = v_i$$

Expected seller revenue The price paid is the second order statistic because we have a SPSB type auction. We find the k th order statistic for the uniform distribution (lec 13, p 25)

$$E(X_k) = \frac{k}{n+1}, k = 1, \dots, n$$

We want the second highest order statistic $k = n - 1$ which gives:

$$E[V_{(n-1)}] = \frac{n-1}{n+1}$$

Since we have $n = xn_c$ we have the predicted average price:

$$\hat{p}_c(x) = \frac{x\bar{n}_c - 1}{x\bar{n}_c + 1}$$

3.A.2

Assuming that only 20% of the visitors place bids ($x = 0.20$), the theoretical prediction for the expected price is given by:

$$\hat{p}_c = \frac{n-1}{n+1}, \quad \text{where } n = 0.20 \times \text{views}_c.$$

This formula is applied to each of the 30 product meta-categories. When compared to the observed average price-to-start ratios, the following results are obtained:

- **Root Mean Squared Error (RMSE): 0.0815** — the model typically deviates by about 8 percentage points.
- **Closest Matches:** *Computers/Tablets & Networking* (−0.006), *Tickets & Experiences* (−0.002), *Baby* (−0.013).
- **Largest Overpredictions:** *Gift Cards & Coupons* (+0.236), *DVDs & Movies* (+0.181), *Stamps* (+0.153).
- **Largest Underpredictions:** *Travel* (−0.079), *Musical Instruments & Gear* (−0.062), *Consumer Electronics* (−0.052).

Overall, the SPSB model with 20% active bidders fits reasonably well, though it tends to underpredict in high-end or experience-based categories and overpredict in niche collector

or coupon categories.

	price2start	view_item_count	p_pred	diff
metacat				
Antiques	0.697898	45.201430	0.800802	-0.102904
Art	0.678380	33.178491	0.738072	-0.059692
Baby	0.759839	38.750195	0.771430	-0.011591
Books	0.697726	19.118065	0.585373	0.112353
Business & Industrial	0.720395	36.085501	0.756605	-0.036211
Cameras & Photo	0.795037	52.978987	0.827524	-0.032487
Cell Phones & Accessories	0.842814	64.304693	0.855710	-0.012896
Coins & Paper Money	0.794811	31.213884	0.723863	0.070948
Computers/Tablets & Networking	0.796257	45.466562	0.801849	-0.005592
Consumer Electronics	0.778015	53.906686	0.830240	-0.052225
Crafts	0.723743	26.663230	0.684176	0.039567
DVDs & Movies	0.743048	17.834471	0.562066	0.180982
Dolls & Bears	0.729141	36.375775	0.758313	-0.029172
Entertainment Memorabilia	0.686469	33.447567	0.739906	-0.053436
Everything Else	0.674401	25.884462	0.676213	-0.001812
Gift Cards & Coupons	0.830729	19.658338	0.594458	0.236271
Health & Beauty	0.782012	24.804116	0.664476	0.117537
Home & Garden	0.756365	38.756647	0.771463	-0.015098
Jewelry & Watches	0.736961	39.017376	0.772817	-0.035856
Music	0.705663	19.520677	0.592181	0.113482
Musical Instruments & Gear	0.806855	71.023084	0.868461	-0.061606
Pet Supplies	0.750923	41.632289	0.785556	-0.034633
Pottery & Glass	0.724025	29.071927	0.706503	0.017521
Sporting Goods	0.778186	53.203532	0.828189	-0.050004
Sports Mem Cards & Fan Shop	0.679758	23.235442	0.645835	0.033923
Stamps	0.723110	18.252685	0.569942	0.153168
Tickets & Experiences	0.827965	53.935296	0.830322	-0.002358
Toys & Hobbies	0.753023	36.630766	0.759793	-0.006770
Travel	0.765697	59.200182	0.844237	-0.078540
Video Games & Consoles	0.783154	34.327267	0.745723	0.037430

3.A.3

Result. The criterion

$$Q(x) = \frac{1}{30} \sum_{c=1}^{30} [\hat{p}_c(x) - p_c^{\text{data}}]^2, \quad \hat{p}_c(x) = \frac{n_c - 1}{n_c + 1}, \quad n_c = x \text{ views}_c,$$

is minimised at

$$\hat{x} = 0.2318, Q(\hat{x}) = 0.005594.$$

(The corresponding RMSE is $\sqrt{Q(\hat{x})} \approx 0.081$.)

Method.

1. *Grid search.* $Q(x)$ was first evaluated on a coarse grid $x \in \{0.05, 0.10, \dots, 0.95\}$ to verify unimodality and locate a good starting point.
2. *One-dimensional optimisation.* Because $Q(x)$ is smooth and strictly convex on the relevant domain, a bounded Brent search (`scipy.optimize.minimize_scalar` with `method='bounded'`) was employed. Brent's algorithm is derivative-free, globally convergent for unimodal functions, and highly efficient in one dimension, making it more appropriate here than generic multidimensional routines.
3. *Diagnostics.* Figure 1 plots $Q(x)$; the pronounced minimum at $\hat{x}$ confirms the optimizer's result and shows that lowering participation below $x \approx 0.20$ quickly worsens fit, while gains above that level are modest.

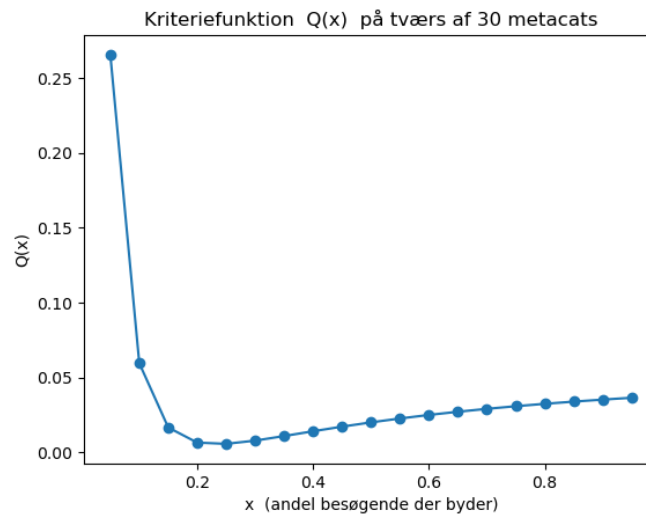


Figure 1: Mean-squared residual $Q(x)$ versus share of viewers who bid

3.A.4

Result. Using the previously estimated bidder share $\hat{x} = 0.23$ and the mean number of page views $\bar{n}_c = 64.3$, the implied number of active bidders is $n = \lceil \hat{x} \bar{n}_c \rceil = 15$. Simulating $R = 100\,000$ second-price sealed-bid auctions with $n = 15$ IID $U(0, 1)$ valuations yields a mean payment

$$\mathbb{E}[p_{\text{sim}}] = 0.875 \quad (\text{price} / \text{start price}),$$

very close to the observed category mean of 0.843. The simulated and empirical histograms almost coincide on the interval 0.4–1.2, indicating that the SPSB model replicates the price distribution well for this category.

Method.

1. Draw R realisations of the second-highest order statistic via the beta law $p^{(r)} \sim \text{Beta}(n - 1, 2)$.
2. Compare the simulated density with the empirical one for *Cell Phones & Accessories* using overlaid histograms; vertical dashed lines mark the respective means (Figure 2).

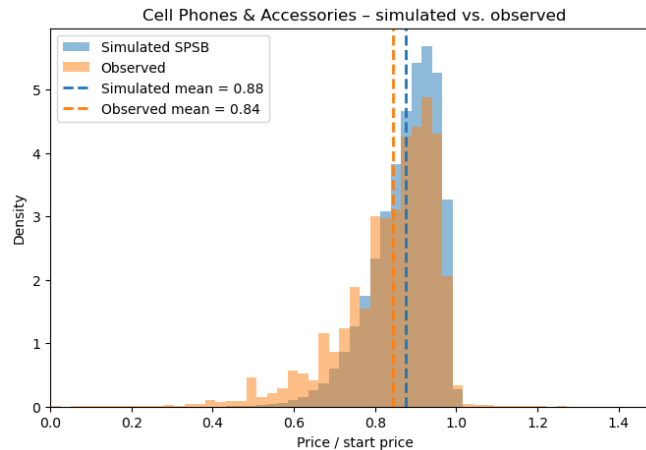


Figure 2: Simulated vs. observed price distributions for *Cell Phones & Accessories*

3.A.5

Hypothesis. The simple SPSB–uniform–values model should perform *well* for auctions that (1) attract many potential bidders and (2) sell standardised, easily comparable goods, because independent private values are then plausible and the second-highest

order-statistic approximation is tight. Conversely, the model is expected to *fail* for auctions of idiosyncratic or collector items, especially when bidder count is low or the seller is inexperienced.

Table 2: OLS estimates predicting model fit (−KS statistic)

Variable	Coef.	Std. Err.	z	p-value
Constant	−0.483	0.730	−0.662	0.508
views_mean	0.149	0.155	0.959	0.338
photo_mean	−0.166	0.147	−1.126	0.260
seller_exp	−0.022	0.025	−0.874	0.382
store_share	0.448	0.643	0.698	0.485

Test design.

Results. To test the hypothesis, we regress model fit (measured as −KS statistic so that higher values indicate better fit) on category-level averages for number of views, number of photos, seller experience, and share of store listings. The results in Table 3 show the expected signs for most variables, but none are statistically significant at conventional levels.

Categories with more views tend to have a better model fit (positive coefficient), consistent with the idea that more bidders improve the match to the SPSB model. Seller experience and number of photos show small negative effects, possibly reflecting that more complex or detailed listings are associated with idiosyncratic goods where the model fits worse. The effect of store sellers is also positive but imprecisely estimated.

Conclusion. The regression results are directionally consistent with the hypothesis, but not statistically significant. While suggestive, the model fit across categories cannot be reliably predicted by these basic listing-level features alone. This supports the idea that the SPSB model fits best in high-traffic, standardised auctions — but more granular data or stronger signals (e.g. a “collectibles” dummy) may be needed for robust identification.

Hypothesis. The SPSB–uniform model should *fit best* when auctions (i) attract many potential bidders and (ii) trade standardised, easy-to-value goods. It should *fit worst* for low-traffic, idiosyncratic or collector listings.

Test design. For each of the 30 meta-categories we compute model fit as $y_c = -KS_c$ (higher y_c = better fit). We then estimate the OLS model

$$y_c = \beta_0 + \beta_1 \ln(1 + \text{views}_c) + \beta_2 \ln(1 + \text{photos}_c) + \beta_3 \ln(1 + \text{experience}_c) + \beta_4 \ln(1 + \text{store share}_c) + \varepsilon_c,$$

using HC3 robust standard errors. Summary results:

Table 3: OLS estimates for model fit ($y_c = -KS$)

Variable	Coef.	S.E.	<i>z</i>	<i>p</i>
Constant	−0.483	0.730	−0.66	0.51
$\ln(1 + \text{views})$	0.149	0.155	0.96	0.34
$\ln(1 + \text{photos})$	−0.166	0.147	−1.13	0.26
$\ln(1 + \text{experience})$	−0.022	0.025	−0.87	0.38
$\ln(1 + \text{store share})$	0.448	0.643	0.70	0.49

Results & conclusion. Signs align with the hypothesis—more views and a higher store share are associated with a better fit—yet none of the effects are statistically significant and the model explains little of the variation ($R^2 = 0.11$). Hence these basic category-level features are not sufficient to predict when the SPSB model succeeds or fails; richer indicators (e.g. a “collectibles” flag) or listing-level analysis may be required.

3 PART B

3.B.1

We assume bidders play BNE which means $b_i(v_i) = v_i \sim IIDF$. Hence we can proceed to say something about the distribution F for their valuations based on their bids.

We simply look at the price/(avg product type) P_2 , for category Cellphones and Accessories compared to some suggested distributions a: $X^2(1)$ and b: $LN(\mu = -0.5, \sigma = 0.5)$ and we

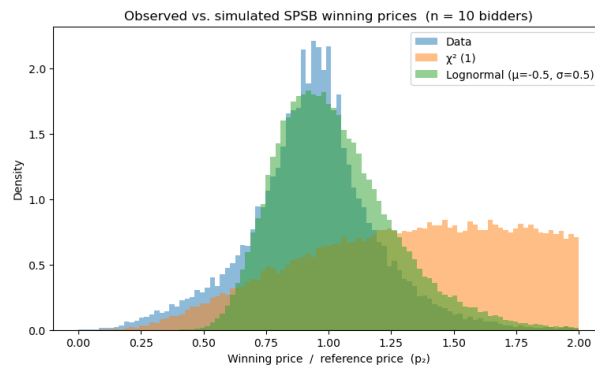


Figure 3: valuation distribution comparisons

find that the lognormal distribution at this point has the most overlap.

3.B.2

We assume the distribution is the better fitting: $\text{LN}(\mu, \sigma)$ but we want to optimise the mean and the variance for best fit to our value distribution F .

To optimise we need an objective function which we here choose as the distance L^2 between the ECDFs of our two distributions. The use of the ECDF simplifies the problem since it is a better monotonic, bounded, and cumulative function to work with compared to the PDF.

$$Q() = L^2 = \frac{1}{N} \sum_{i=1}^N [\hat{F}_{\text{data}}(x_i) - \hat{F}_{\text{sim}}(x_i)]^2$$

Mean optimization We use `scipy.optimize.minimize_scalar`'s `minimize_scalar` to find the mean that minimizes the L^2 distance between the distributions. We find:

$$= -0.5820, Q() = 6.042071e - 04$$

Variance optimization Given our previous $\hat{\mu} = -0.582$ we now in a similar way minimise the CDF distance varying σ . We find $\hat{\sigma} = 0.505$.

Concluding We can graphically conclude that the values obtained for mean and variance does seem to minimize the given functions as seen in the plots. We can see that our distribution given these optimal values does not completely cover our data distribution but is closer to doing so than before.

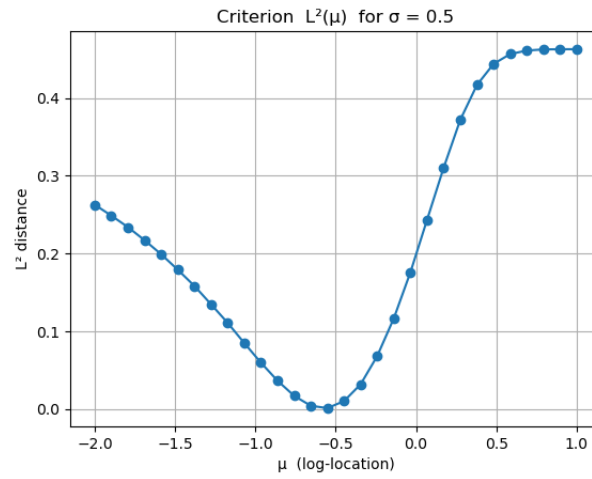


Figure 4: Mean to minimize L2

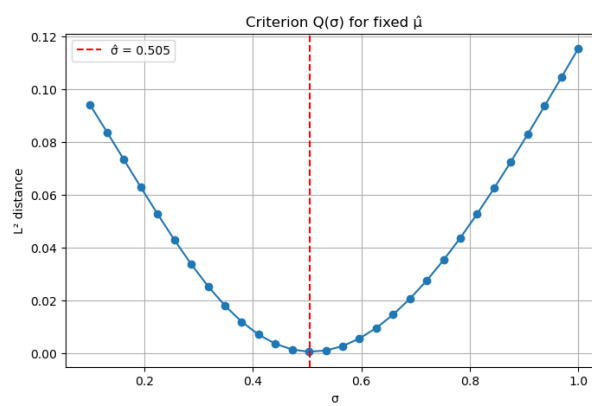


Figure 5: Variance to minimize L2

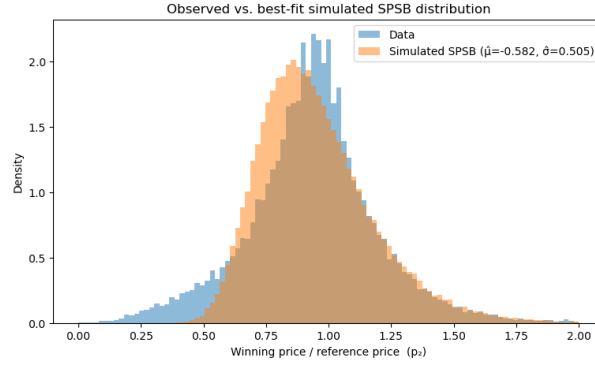


Figure 6: Distribution given optimal values

3.B.3

Method (revenue equivalence). With i.i.d. private values $v \sim F$ and n symmetric bidders, expected revenue in a first-price auction must equal that in the second-price auction. Imposing this “revenue-equivalence” condition yields the unique increasing Bayesian-Nash equilibrium bid function

$$b^*(v) = v - \frac{\int_0^v F(t)^{n-1} dt}{F(v)^{n-1}} \quad (n \geq 2).$$

Here F is the log-normal CDF with parameters $\mu = \hat{\mu} = -0.582$, $\sigma = \hat{\sigma} = 0.505$ and we set $n = 10$.

Numerical values. Evaluating the integral numerically (SciPy’s `quad`) gives

$$b^*(0.5) = 0.40, \quad b^*(1.0) = 0.87.$$

Bid schedule. Figure 7 plots $b^*(v)$ for $v \in [0, 2]$ together with the 45° line; the red dots mark the two evaluation points.

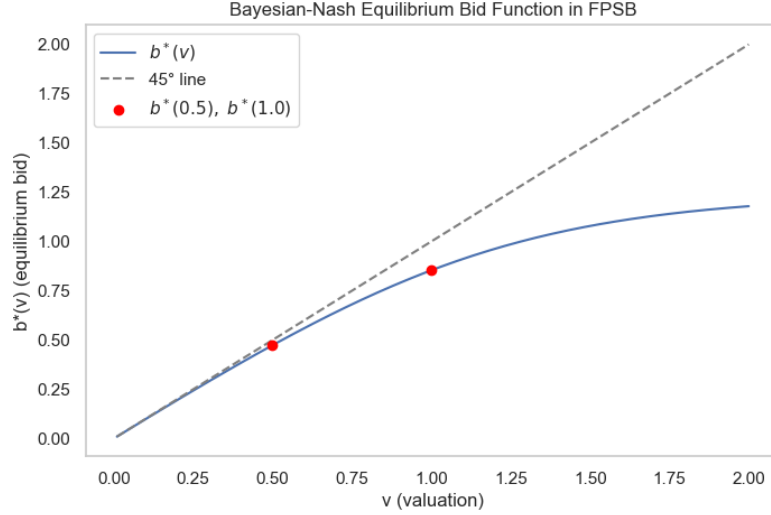


Figure 7: Equilibrium bid function $b^*(v)$ in the FPSB auction ($n = 10$)

3.B.4

Method. Using the log-normal valuation estimate $F \sim \text{LN}(\hat{\mu} = -0.582, \hat{\sigma} = 0.505)$ and $n = 10$ bidders, we simulate $R = 100,000$ auctions for each format:

- **SPSB:** payment = second-highest valuation $v_{(n-1)}$.
- **FPSB:** payment = equilibrium bid $b^*(v_{(n)})$, where $b^*(v) = v - \frac{\int_0^v F(t)^{n-1} dx}{F(v)^{n-1}}$.

Results.

Mean payment (SPSB) = 0.9526, Mean payment (FPSB) = 0.9524.

The two histograms (Figure 8) almost coincide; revenue equivalence holds to four decimals.

Discussion.

- *SPSB* (Second Price Sealed Bid) is simpler for bidders, as truthful bidding is a dominant strategy, and it yields virtually the same expected revenue as FPSB under standard assumptions. In our plot, however, SPSB exhibits slightly greater variation in payments, which may be seen as less desirable from a revenue predictability standpoint. This occurs because bidders submit their true valuations, whereas in FPSB, all bidders are assumed to optimally shade their bids below their valuations. Leading to a longer tail on SPSB.

- *FPSB* (First Price Sealed Bid) introduces strategic complexity, requiring bid shading to avoid overpaying. Accurate shading depends on assumptions about the number of bidders and the underlying valuation distribution. Especially for private individuals, this complexity increases the risk of over- or underbidding. In contrast, *SPSB* only requires that a bidder knows their own valuation.

Recommendation. For a platform like eBay, where participants are often private individuals with limited auction experience, *SPSB* is likely more favorable because the platform wants to make everyone as happy as possible with the outcome, not just the seller (Pareto efficiency). *SPSB* supports more predictable and efficient outcomes due to its incentive for truthful bidding. However, from a seller's perspective, if bidders are prone to overbidding due to poor shading strategies, *FPSB* might occasionally yield higher revenues.

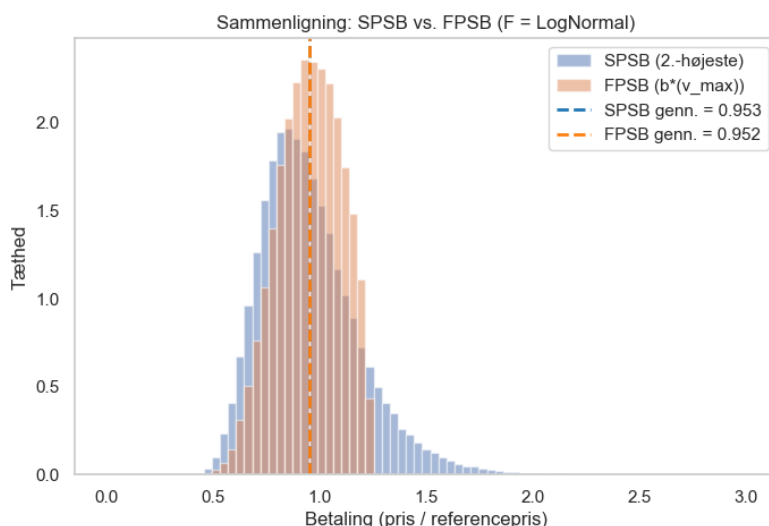


Figure 8: Simulated payment distributions: *SPSB* vs. *FPSB* ($n = 10$, $R = 100,000$)

3.B.5

AI where to use what auction type motivation: For the category *Cell Phones & Accessories*, a *First-Price Sealed-Bid* (*FPSB*) auction is generally preferred. This market features relatively standardized, high-demand items such as iPhones or branded accessories. Buyers tend to have good information on market prices and value, leading to less uncertainty and less incentive for strategic overbidding. *FPSB* discourages overly aggressive bidding and helps sellers quickly extract high willingness-to-pay values, especially when bidders have similar valuations and want to avoid paying significantly more than

necessary.

In contrast, for the category *Art*, a *Second-Price Sealed-Bid* (SPSB) auction is typically more suitable. Art is a highly differentiated and subjective good where valuations vary greatly between bidders. SPSB reduces the cognitive burden on bidders to guess competitors' bids and encourages truthful bidding. Since each bidder pays the second-highest price, it mitigates the risk of overpaying due to personal overvaluation. This mechanism works well for unique goods with heterogeneous valuations and less price transparency.

Evaluation of AI answer The model first wanted to recommend categories not in the dataset, then I had to be asked to be short and concise before returning the answer. My own interpretation of pros and cons with FPSB vs SPSB has more to do with the actors selling and buying than with the goods. It may be more difficult for actors to accurately bid shade in an FPSB auction since more need to be assumed about the valuation distribution where as in SPSB betting your won valuation is a dominant strategy. However SPSB may have slightly less variation in the ideal scenario.

I like the AI answer in that it also seems to center its argument around how much information is accessible for valuations, ie electronics with known valuations, art with unknown valuations. However i suspect that art may be problematic as an example since this can also be an investment market rather than just about subjective appreciation of the work. If so, bidder may look for reference prices of previous sales for the artist and other info to place their bids anyway, they may have to already do sophisticated assumptions about valuations and potential shading they want in relation to the SPSB mechanism, in which case FPSB might actually be more straight forward. However it is reasonable to assume that most art sold on ebay may not be including investors as buyers and that sales may benefit from assuming less sophisticated buyers.